AI 101 for Mechanical Engineers · Module 10

Protect the evaluation boundary

Lesson 10.1

What you will learn

> Distinguish training, validation and final-test data, explain cross-validation, and choose an evaluation boundary that matches the intended engineering use.

Lesson overview

Module 10 · Lesson 1

Protect the evaluation boundary

AI 101 for Mechanical Engineers

1 Give each dataset a clear role

Training	Fit model parameters and learned preprocessing.
Validation	Choose models, settings and alert thresholds.
Final test	Assess the locked procedure after development choices are fixed.

If test results guide more changes, the test becomes development data. A filename alone does not protect the boundary.

2 Cross-validation stays in development

Repeat training and validation across folds of the development data.



train validate · 100 examples, 5 folds of 20: each round trains on 80 and validates on 20

Fit scalers and imputers inside each training fold; transform the validation fold with those stored values. Stratify class proportions only within valid boundaries.

3 Match the split to intended use

Forecasting	Respect time order
Unseen equipment	Hold out whole machines
Independent designs	Random row splits may suit

Overlapping windows: 5,000 windows from 10 machines, split randomly by row, put every machine in both sets. For unseen-machine evaluation, keep whole machines and related events together.

Many rows do not prove independence.

Key takeaway **Define what unseen means for the engineering question before choosing a split.**

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Define “unseen” through the deployment question

- Generalization: performance on examples outside model development
- Later observations and absent machines are different questions
- Each needs a different evaluation boundary
- A large number of rows does not prove independence

Three datasets have different functions

Dataset	Role
Training	Fits parameters, e.g. regression coefficients
Validation	Guides choices: model family, tree depth, threshold
Final test	Evaluates the locked procedure

A test set whose results guide tuning becomes development data. Fix model, preprocessing and choices first.

Cross-validation stays inside development data



- Repeats training and validation over different folds [39]
- Assumes independence and equal fold sizes
- Does not replace a reserved final test

Fit learned preprocessing inside each training fold

1. Form the folds
2. Fit the scaler or imputer on the training portion
3. Transform the validation portion with those fitted values
4. Repeat separately in every iteration

Stratify only within valid boundaries

- For exchangeable classification examples, stratification can keep approximate class proportions
- It does not replace machine, event or time boundaries
- Choose the split design first, then preserve class balance where feasible

Match the split to the deployment question

Question	Split
Forecasting	Respect chronological order
Transfer to new equipment	Hold out whole machines
Independent experimental designs	Randomized row splits may be justified

> There is no universal split rule for every engineering dataset [54].

Overlapping windows can make results optimistic

Random row split

- Each machine’s windows in both sets
- Sets share event information
- Does not test unseen machines

Corrected arrangement

- Hold out whole machines
- Keep overlapping events within consistent boundaries
- Fit preprocessing on training; lock, then evaluate

Teaching example: 5,000 overlapping windows from ten machines; not a required dataset size.

Choose the boundary before the API

- train_test_split and StratifiedKFold: common tools for exchangeable examples
- They do not replace time-ordered or group-based holdouts
- Keep same-machine, same-event or future examples together when required

CHECK YOUR UNDERSTANDING

Which split would assess transfer to a machine never used in development?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Which split would assess transfer to a machine never used in development?

> Hold out whole machines and evaluate the locked procedure on them. Mixing each machine’s rows across development and test assesses a different question.

CHECK YOUR UNDERSTANDING

In the five-fold example, why does each iteration use 80 training examples?

Answer before moving on.

CHECK YOUR UNDERSTANDING

In the five-fold example, why does each iteration use 80 training examples?

> The 100 independent examples form five equal groups of 20. One group is held out for validation; the other four give 80 training examples.

Key takeaways

- Training fits; validation guides choices; final test evaluates the locked procedure
- Keep cross-validation in development; learned preprocessing in training folds
- Define unseen examples through the intended deployment question
- Row counts do not prove independence; check machine and event overlap

References

[39] S. Arlot and A. Celisse, “A survey of cross-validation procedures for model selection,” Statist. Surv., vol. 4, pp. 40–79, 2010. doi: 10.1214/09-SS054

[54] D. R. Roberts et al., “Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure,” Ecography, vol. 40, no. 8, pp. 913–929, 2017. doi: 10.1111/ecog.02881

Numbers match the course reference list. The written lesson is the authoritative text.